DIAGONALS OF MORPHISMS BETWEEN AFFINES ARE CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4243-4252, following the statement at 4238-4241.

Editorial note. The identification of the diagonal with Spec of the multiplication map is the standard affine diagonal fact; the concluding engine example-closed-immersion-affines is named.

Allowed standard prerequisites (editorial).

- Closed immersions of affines from ideals. example-closed-immersion-affines (same chapter, cited at ~4251). Concluding identification of the diagonal as a closed immersion.

Source reference. Original reference 01IG: sources/stacks-schemes/pinned-original-source.tex, source lines 1794–1794. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The diagonal morphism of a morphism between affines is closed.*

Proof. The diagonal morphism associated to the morphism $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is the morphism on spectra corresponding to the ring map $S \otimes_R S \to S$, $a \otimes b \mapsto ab$. This map is clearly surjective, so $S \cong S \otimes_R S/J$ for some ideal $J \subset S \otimes_R S$. Hence Δ is a closed immersion according to Example 01IG. □

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